Hong Nguyen

Curriculum Vitae

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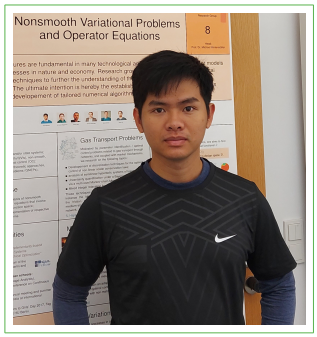

"The essence of mathematics lies in its freedom." - Georg Cantor
"It is true that a mathematician who is not somewhat of a poet, will never be a perfect mathematician." - Karl Weierstrass

Education

2020–now **PhD student enrolled in Humboldt-Universität zu Berlin**, *Weierstrass Institute for Applied Analysis & Stochastics* (WIAS), Berlin, Germany.
In the Research Group Nonsmooth Variational Problems & Operator Equations, Head: Prof. Michael Hintermüller, within the European Innovative Training Network Reduced Order Modelling, Simulation & Optimization of Coupled Systems (ROMSOC), in the Project 11 Optimal Shape Design of Air Ducts in Combustion Engines.

2018–2019 **Master de Sciences, Technologie, Santé, Mathematiques et Applications**, *Université de Rennes 1*, Rennes, France, *assez bien*.
Master 2 in Fundamental Mathematics & Applications within ENS Rennes program (skipped Master 1).

2014–2018 **Bachelor of Science in Mathematics**, *University of Science* (HCMUS), *Viet Nam National University, Ho Chi Minh City* (VNU HCM), Vietnam, *very good*.
Specialized in Mathematical Analysis & Numerical Analysis.

2011–2014 **Elementary Mathematics**, *Ben Tre High School for Gifted Students* (CBT), Vietnam, *very good*.

Master Thesis

Title *Staggered & well-balanced discretization of shallow-water equations*. [pdf][slide]

Supervisor Prof. Nicolas Seguin

Description This thesis investigates some staggered well-balanced finite volume schemes for the shallow water equations. A new staggered scheme, called "offset equilibrium", is developed to conquer the *dry-wet transition* issues where almost all the traditional finite volume schemes collapse. Higher-order extensions for this staggered scheme are also mentioned.

Master Courses

2018–2019 **Courses of Master 2 Program in Fundamental Mathematics with Applications, 2018–2019:**

Sep 2018 Prof. Nicoletta Tchou, Sobolev spaces & elliptic equation.

Sep 2018 Prof. Florian Méhats, Hyperbolic equations.

Nov 2018 Prof. Nicolas Seguin & Eric Darrigrand, Finite Element Method (FEM).

Nov 2018 Prof. Nicolas Seguin, Numerics of transport.

Nov 2018 Prof. Mohammed Lemou, Numerical methods for kinetic transport equations.

Dec 2018 Prof. Luis Miguel Rodrigues, Master 2 Seminar
▷ Hong Nguyen. "On the smallest constant for a Gagliardo-Nirenberg functional inequality". [pdf][slide].

Mar 2019 Vincent Duchêne, Modeling & mathematical analysis of shallow free-surface fluid flows.

Mar 2019 Prof. Marc Quincampoix, Optimal control & Hamilton-Jacobi equations.

Research Interests

- **Mathematical Analysis** ▷ Partial Differential Equations (PDEs), Differential Geometry (DG).
- **Numerical Analysis** ▷ Finite Difference Methods (FDMs), Finite Element Methods (FEMs), Finite Volume Methods (FVMs).
- **Mathematical Optimization** ▷ Optimal Control, Shape Optimization, Topology Optimization.
- **Fluid Mechanics** ▷ Fluid Dynamics, Computational Fluid Dynamics (CFDs).

○ **Turbulence** ▷ Turbulence modeling.

Experience

Vocational

Apr 2021–now **Software Development Engineer (SDE)**, *Math$_+$ Tec GmbH*, Vienna, Austria.
Secondments for technical & scientific training at the industrial partner within ROMSOC project.

Mar–Jun 2019 **Master Internship**, IRMAR, Rennes, Bretagne, France.

Miscellaneous

2015–2017 **Olympiad Trainer**, *Ben Tre High School for Gifted Students* (CBT), Vietnam.
Trained the selected team of CBT for Vietnamese Mathematical Olympiads (VMOs).

Publications

[1] Nguyen Anh Dao, Jesús Ildefonso Díaz, Quan Ba Hong Nguyen. "Pointwise Gradient Estimates in Multi-dimensional Slow Diffusion Equations with a Singular Quenching Term". *Adv. Nonlinear Stud.* (2020). DOI https://doi.org/10.1515/ans-2020-2076. [pdf]

Courses

Oct 12–23, 2020 Thematic Einstein Semester (TES): Energy-based mathematical methods for reactive multiphase flows.
▷ TES Student Compact Course: *Variational Methods for Fluids & Solids*.

Apr–Jun 2020 Dr. Stephan Schmidt. *Computational Geometry & Shape Optimization*. Master course, Humboldt Universität zu Berlin, 2020.

Seminars

Mar 2020–now Berliner Oberseminar Nichtlineare partielle Differentialgleichungen (Langenbach-Seminar), Research Group 1: *Partial Differential Equations*, WIAS Berlin, Germany.

Mar 2020–now Material Modeling Seminar, WIAS Berlin, Germany.

Mar 2020–now Numerical Mathematics Seminars, Research Group 3: *Numerical Mathematics & Scientific Computing*, WIAS Berlin, Germany.

Mar 2020–now Modern Methods in Applied Stochastics & Nonparametric Statistics, Research Group 6: *Stochastic Algorithms & Nonparametric Statistics*, WIAS Berlin, Germany.

Mar 2020–now Joint Research Seminar on Mathematical Optimization, Research Group 8: *Non-smooth Variational Problems & Operator Equations*, WIAS Berlin, Germany.

Mar 2021–now Optimization, Control & Inverse Problems, Research Group 8: *Non-smooth Variational Problems & Operator Equations*, WIAS Berlin, Germany.

Mar 2020–now Berlin Mathematics Research Center MATH$^+$ Events.

Nov 2020–Apr 2021 ROMSOC Scientific ESRs Seminar.

Schools

Sep 7–10, 2020 Model Order Reduction Summer School 2020 (MORSS2020), École polytechnique fédérale de Lausanne (EPFL), Lausanne, Switzerland.

Jul 20–24, 2020 162nd European Study Group with Industry, University of Leeds, UK.

Workshops

Apr 29, 2021 FUHRI2021: Finite volUme metHods for Real-world applIcations, WIAS Berlin, Germany.

Sep 28–29, 2020 ROMSOC Internal Workshop.

Sep 15, 2020 ROMSOC Ethical Monitoring - 3rd Workshop on Ethics.

Conferences

Mar 24–25, 2021 Annual Meeting of the Second Phase of the DFG Priority Programme 1962 (SPP1962): Non-smooth & Complementarity-based Distributed Parameter Systems: Simulation & Hierarchical Optimization. Deutsche Forschungsgemeinschaft (DFG) + WIAS, Germany.

Sep 14–17, 2020 Deutsche Mathematiker-Vereinigung (German Mathematical Society) Annual Meeting 2020 (DMV2020), Technische Universität Chemnitz, Chemnitz, Germany.

Talks

Apr 12, 2021 Hồng Quản Bá Nguyễn (WIAS Berlin). "*Optimal Shape Design of Air Ducts in Combustion Engines: Design a General Shape Optimization Framework*". ROMSOC ESR Seminar. [abstract]

Sep 7, 2020 Hong Nguyen (WIAS Berlin). "*A shape optimization problem for stationary Navier-Stokes flows in three-dimensional tubes*". Model Order Reduction Summer School 2020 (MORSS2020), École polytechnique fédérale de Lausanne (EPFL), Lausanne, Switzerland. [slide]

Jun 26, 2019 Hong Nguyen. "*Staggered & well-balanced discretization of shallow water equations*". PhD interview, Department of Mathematics, RWTH Aachen University, Aachen, Germany. [slide]

Miscellaneous

Feb 22, 2021 WIAS-Day2021 ▷ Hong Quan Ba Nguyen (Research Group 8)'s Research Profile.

Nov 30, 2020 ROMSOC Blog ▷ Hong Nguyen. *Optimal shape design of air ducts* [pdf].

Visits

Apr 2021–now MATH+TEC GmbH, Vienna, Austria.

Apr 2021–now Vienna University of Technology/Technische Universität Wien, Vienna, Austria.

Apr 23, 2021 University of Vienna/Universität Wien, Vienna, Austria.

Mar 2020–now Humboldt-Universität zu Berlin, Berlin, Germany.

Jun 25–27, 2019 Department of Mathematics, Rheinisch-Westfälische Technische Hochschule (RWTH) Aachen University, Aachen, Germany.

Sep 2018–Jun 2019 UFR Mathématiques, Université de Rennes 1, Rennes, Bretagne, France.

Sep 2018–Dec 2018 Université Rennes 2 - UFR Sciences sociales, Rennes, Bretagne, France.

Mar 2017 Vietnam Institute for Advanced Studies in Mathematics (VIASM), Hanoi, Vietnam.

Awards & Scholarships

Mar 2020–Aug 2021 Marie Skłodowska-Curie Actions (MSCA) PhD Fellowship, *Horizon 2020 - Innovative Training Network - European Industrial Doctorate Programs*, Grant Agreement number: 765374 - Reduced Order Modelling, Simulation & Optimization of Coupled Systems (ROMSOC) [H2020-MSCA-ITN-2017-EID/ROMSOC/ESR11].

2018–2019 Lebesgue Master Scholarship for *Master 2* program within ENS Rennes program.

2017 Scholarship of *Vietnam Institute for Advanced Studies in Mathematics* (VIASM).

2016 Special Prize ⇐ First Prize in Algebra + First Prize in Analysis, in *Vietnamese Mathematical Olympiad for College Students* (VMC 2016).

2015 Second Prize in Algebra, Second Prize in Analysis, in *Vietnamese Mathematical Olympiad for College Students* (VMC 2015).

2012 Gold Medal ranked #2, in *Vietnamese Apr 30th Mathematical Olympiad for Central & Southern Vietnam*.

Computer Skills

GitHub GitHub/NQBH
▷ GitHub/ROMSOC project.

Basic R, Fortran, Fireshape, Firedrake, FEniCS, Python 3, STAR-CCM+.

Intermediate git, C/C++, OpenFOAM.

Advanced UNIX/Linux [openSUSE, (K)Ubuntu], TEX/LATEX, MATLAB.

Languages

Vietnamese Mother tongue: *Literature-oriented*

English A *fluent speaker/reader* & an *advanced listener/writer*

Referees

[5] **Prof. Michael Hintermüller** *Primary PhD Supervisor*

[4] **Prof. Nicolas Seguin** *Master Supervisor*

[3] **Dr. Nguyen Anh Dao** *Bachelor Supervisor*

[2] **Dr. Tan Trung Nguyen** *Bachelor Teacher*

[1] **Dr. Manh Khang Dao** *Bachelor Teacher*

Hobbies

- Psychology
- Philosophy
- Literature
- Blogging
- Arts (emotions carriers): movies, animes, mangas, (instrumental) musics, etc. □

Updated to Apr 28, 2021.